

TOI-1811b

R-2047

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_260328 · created 2026-08-08T23:48:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461127.7139 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.02
Transit depth (Rp/Rs)^2	2.41 +/- 0.62 [%]
Orbital Inclination (inc)	86.74 +/- 1.0
Airmass coefficient 1 (a1)	0.985 +/- 0.015
Airmass coefficient 2 (a2)	0.012 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.56 %
Best Comparison Star	#3 - [133, 325]
Optimal Method	PSF photometry
Transit Duration (day)	0.08 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.155 ± 0.02 vs 0.13272 (deviation 0.87σ)
Duration fit vs published	0.08 d vs 0.0813 d (deviation 0.06σ)
Mid-time O–C	3.5 min
Depth SNR	6.1
Residual scatter	1.56 %

Light curve

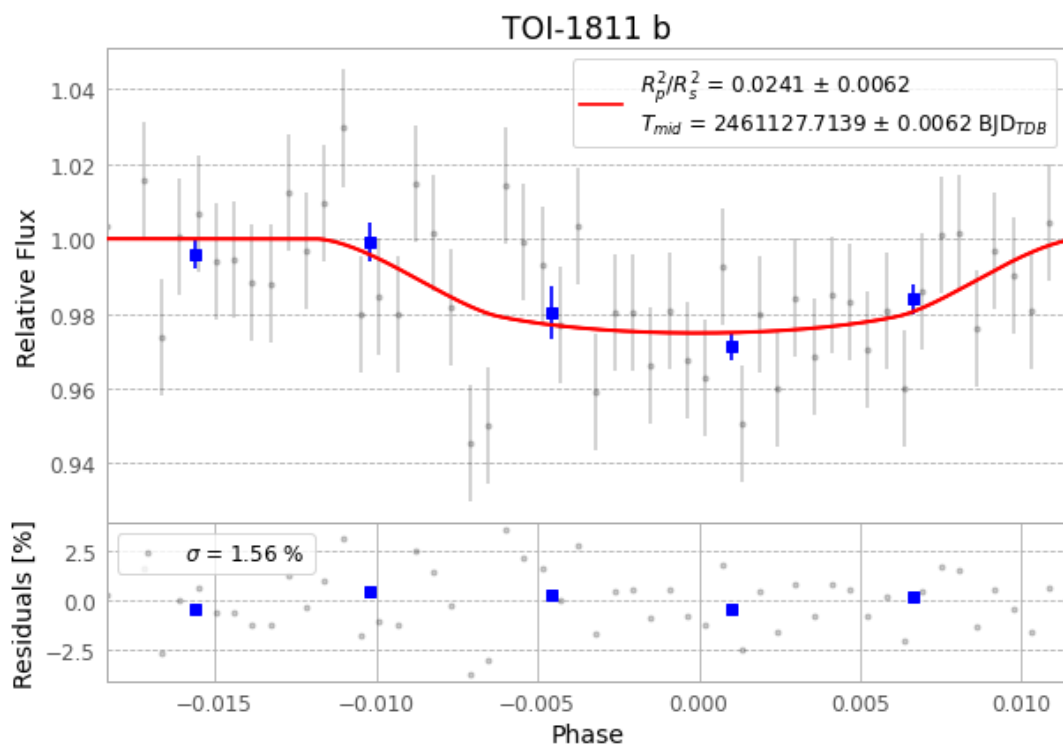


Figure 1 — FinalLightCurve_TOI-1811 b_2026-03-27.png (SHA-256 3e5d2611d2861216...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 941e496c85377c97...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

53 FITS frames (35 MB), TOI1811260328032112.FITS → TOI1811260328060015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-260328.log (7ee16e4a8cf4...), FinalLightCurve_TOI-1811 b_2026-03-27.png (3e5d2611d286...), FinalLightCurve_TOI-1811 b_2026-03-27.csv (9942b10f1d24...)

Compute resources

Wall time 379.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 23:48Z.